

Air Quality Risk Prediction Using Environmental and Meteorological Data

Final Presentation

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QM640: Data Analytics Capstone

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Executive Summary



This capstone develops a predictive framework for the binary classification of poor air quality days in Kapolei, Honolulu County, Hawaii.



The study uses Daily EPA air quality data and NOAA meteorological data for January 1, 2024 to December 31, 2025.



The final cleaned dataset contains 731 daily observations and includes pollutant, weather, AQI, and time based variables.



Poor air quality was operationalized as `'overall_aqi > 35'`, identifying 150 poor air quality days and 581 non-poor air quality days.



Preliminary results show that Random Forest performed best, with strong cross-validated performance and ozone and PM2.5 emerging as key predictors.

Gap Analysis

Air quality prediction is complex because pollutant levels are influenced by changing environmental, meteorological, and seasonal conditions.

Prior studies support the use of pollutant and weather variables for AQI prediction, but model performance and interpretability remain important concerns.

The study addresses this gap by combining EPA pollutant data, NOAA weather data, seasonal features, and comparative machine learning models for Kapolei, Hawaii.

Many studies focus on broader AQI forecasting, while this capstone focuses specifically on daily binary classification of poor air quality days.

The practical gap is the need for a simple, interpretable framework that can support local monitoring, risk awareness, and sustainability planning.

Research Questions

Research Question	Null and Alternative Hypothesis	Method Used	Test Result
RQ1: Which environmental and meteorological variables are most strongly associated with poor air quality days?	H₀₁ : Environmental and meteorological variables are not significantly associated with poor air quality days. H_{a1} : At least one environmental or meteorological variable is significantly associated with poor air quality days.	Correlation analysis and Random Forest feature importance	Ozone and PM2.5 were the strongest preliminary factors.
RQ2: Can poor air quality days be accurately identified and classified using pollutant and weather data?	H₀₂ : Poor air quality days cannot be classified using pollutant and weather data better than chance. H_{a2} : Poor air quality days can be classified using pollutant and weather data significantly better than chance.	Binary classification models	Yes. Models classified poor air quality days with strong preliminary performance.
RQ3: Do poor air quality days vary significantly across seasons and time periods?	H₀₃ : There are no statistically significant differences in poor air quality days across seasons and time periods. H_{a3} : Poor air quality days differ significantly across seasons and time periods.	Seasonal/monthly EDA; inferential testing pending	Winter and spring showed higher AQI and more poor air quality days.
RQ4: Which predictive model performs best for binary classification?	H₀₄ : There is no significant difference in predictive performance among the selected models for the binary classification of poor air quality days. H_{a4} : At least one predictive model performs significantly better than the others for the binary classification of poor air quality days.	Model comparison using F1-score and ROC-AUC	Random Forest best overall.

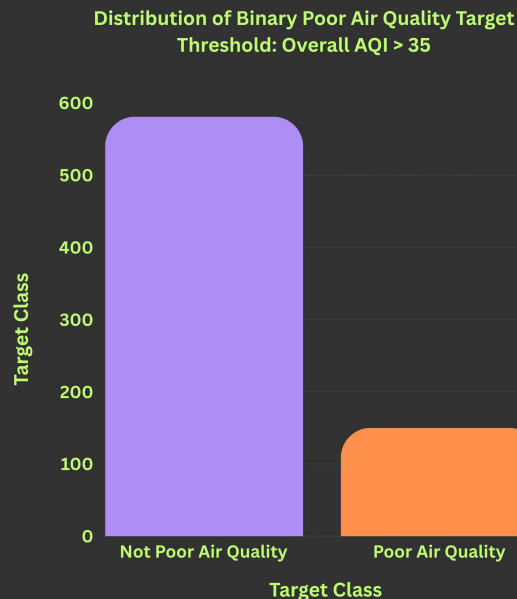
Data Description and EDA

Data Description

- **Study setting:** Kapolei, Honolulu County, Hawaii
- **Study period:** January 1, 2024 to December 31, 2025.
- **Unit of analysis:** Daily observation
- **Final dataset:** 731 rows and 41 variables
- **Data sources:** EPA AirDATA / AQS daily pollutant files and NOAA CDO daily weather summaries

Key Variables

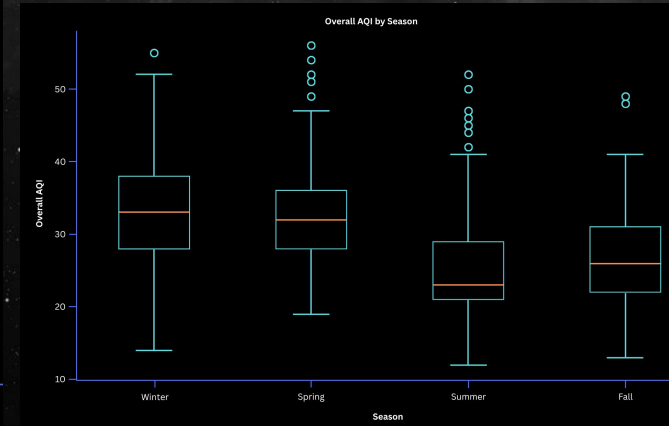
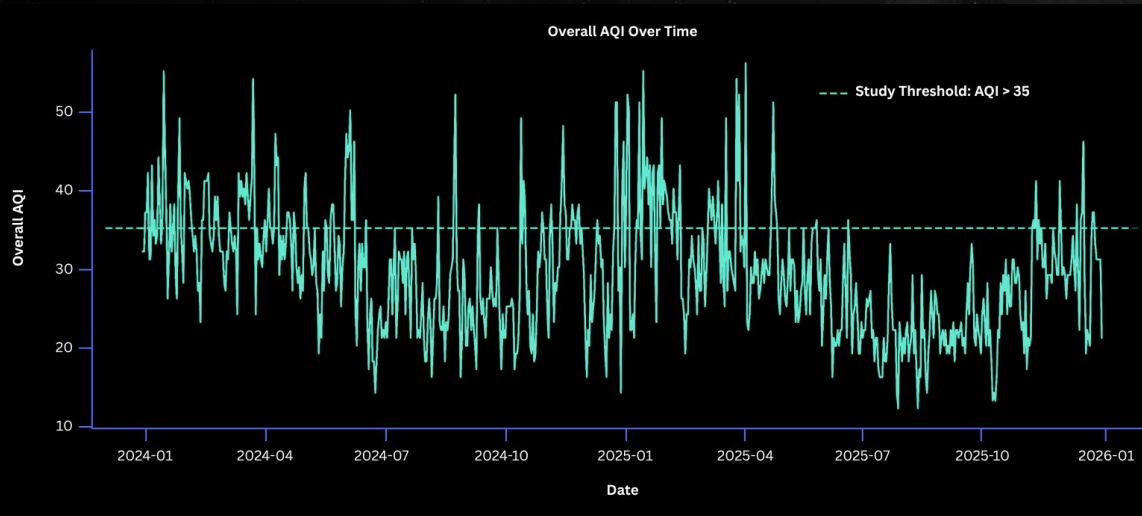
- **Pollutants:** ozone, PM2.5, PM10, NO2
- **Weather:** maximum temperature, minimum temperature, precipitation, average wind speed
- **Time features:** year, month, season, day of week, weekend indicator
- **Target:** 'poor_air_quality', derived from 'overall_aqi'



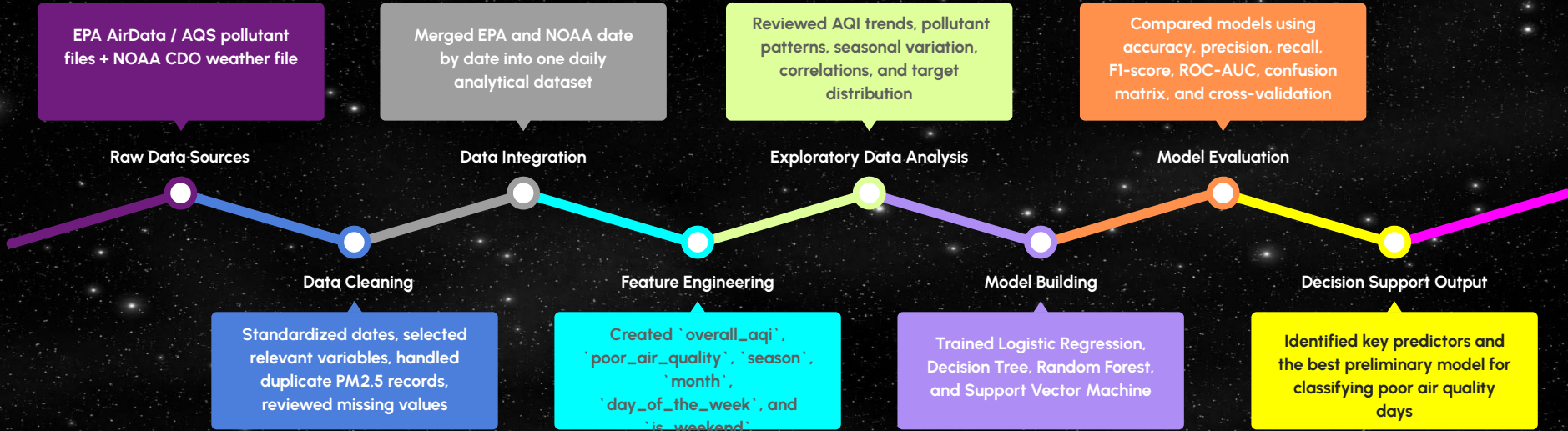
Note: To address the remaining class imbalance, as seen in the distribution above, SMOTE will be tested as a modelling enhancement on the training data before final model selection.

EDA Findings

- 'poor_air_quality' was defined as 'overall_aqi > 35'
- Winter and spring showed the highest mean overall AQI values and the most poor air quality days.
- Ozone was the dominant pollutant on 474 days, followed by PM2.5 on 256 days.
- The final target distribution was 150 poor air quality days and 581 non-poor air quality days.



Architecture/Diagram Workflow



Model Building

Target Variable

- `'poor_air_quality'`
- Binary target derived from `'overall_aqi'`
- Final study threshold: `'overall_aqi > 35'`

Predictor Features

- Pollutants: ozone, PM2.5, PM10, NO2
- Weather: `'tmax'`, `'tmin'`, `'prcp'`, `'awnd'`
- Time features: `'month'`, `'is_weekend'`

Candidate Models

- Logistic Regression: interpretable baseline model
- Decision Tree: interpretable nonlinear model
- Random Forest: ensemble model with feature importance
- Support Vector Machine: nonlinear margin-based classifier

Validation Approach

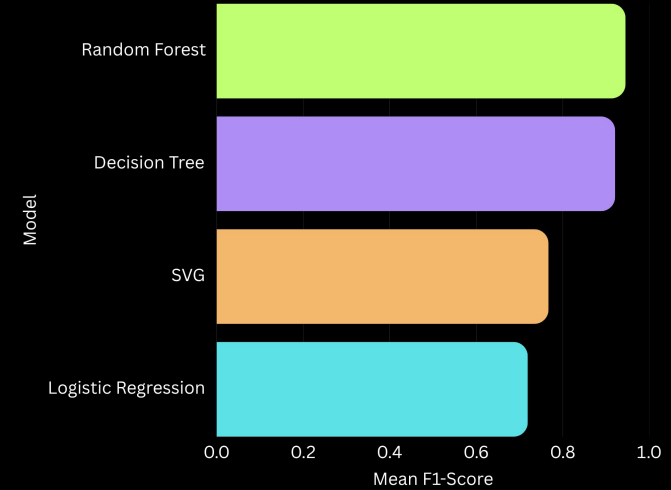
- 80/20 stratified train-test split
- Stratified 5-fold cross-validation
- Evaluation metrics: accuracy, precision, recall, F1-score, and ROC-AUC

Best Preliminary Model: Random Forest

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
2	Random Forest	0.986	0.967	0.967	0.967	0.994
1	Decision Tree	0.973	0.906	0.967	0.935	0.992
3	SVG	0.918	0.737	0.933	0.824	0.976
0	Logistic Regression	0.878	0.658	0.833	0.735	0.934

- Random Forest had the strongest overall performance among the four candidate models.
- Test set performance:
 - Accuracy: 0.986
 - Precision: 0.967
 - Recall: 0.967
 - F1-score: 0.967
 - ROC-AUC: 0.994

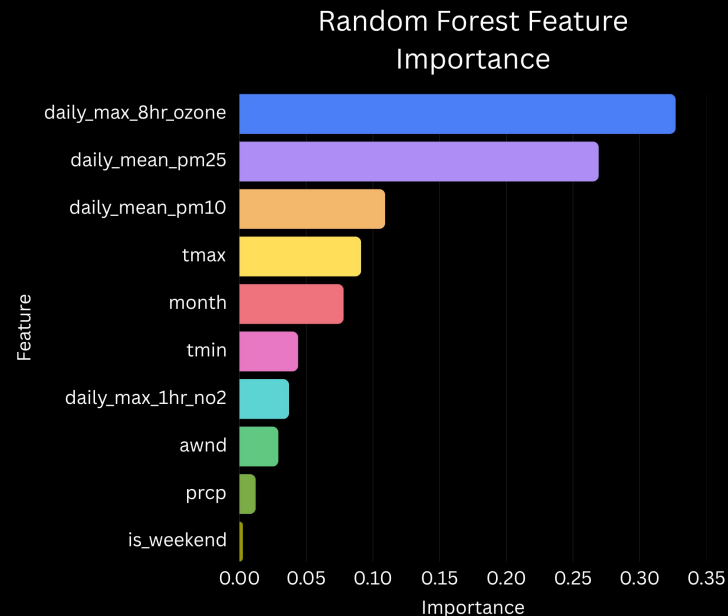
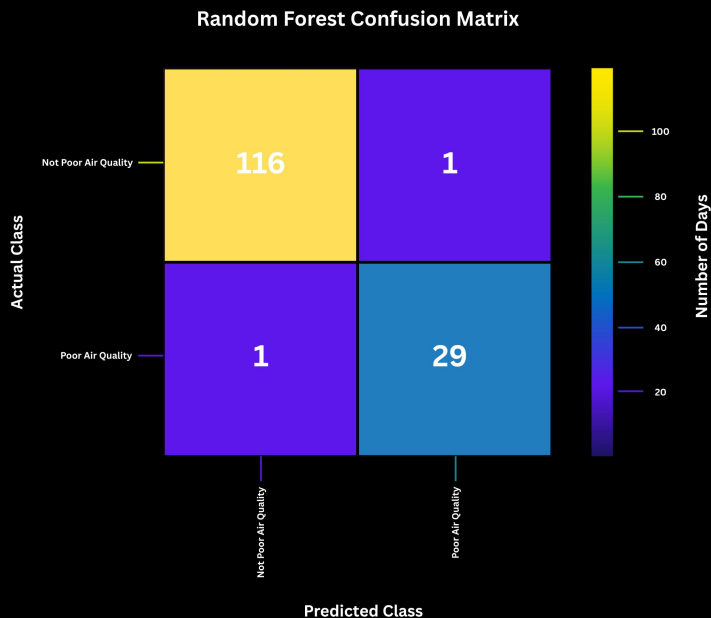
Cross-Validation Results



- Random Forest remained the best model under stratified 5-fold cross-validation.
- Cross-validated mean F1-score: 0.945
- Cross-validated mean ROC-AUC: 0.991

Model Interpretation

- Random Forest confusion matrix: 116 true negatives, 29 true positives, 1 false positive, and 1 false negative.
- Feature importance showed that ozone and PM2.5 were the strongest predictors.
- Results suggest that poor air quality days can be classified effectively using pollutant, weather, and time based variables.



Implementation and User Benefit

- Supports local environmental monitoring by identifying days with elevated air quality risk.
- Helps sustainability teams and planners understand which pollutant and weather conditions are associated with poor air quality days.
- Provides a foundation for a simple dashboard or decision support tool using daily pollutant and weather inputs.
- Improves communication by translating multiple pollutant and weather measures into one binary risk classification.
- Supports proactive planning by moving from retrospective AQI review toward earlier risk awareness.

Conclusion

- This study developed a daily analytical framework for classifying poor air quality days in Kapolei, Hawaii.
- EPA pollutant data and NOAA weather data were successfully cleaned, merged, and used for EDA and preliminary modeling.
- Ozone and PM2.5 emerged as the strongest preliminary predictors of poor air quality classification.
- Random Forest was the best performing preliminary model based on test set and cross-validation results.
- Final work will include additional inferential testing, final model refinement, limitations, and recommendations for practical environmental monitoring use.
- The final report will include SMOTE testing, formal seasonal inference, and final model refinement.

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